

Data Preprocessing Assignment

Dataset Information

You will use the **Ames Housing Dataset** from the Kaggle competition "House Prices: Advanced Regression Techniques."

Download link:

<https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques/data>

Download the `train.csv` file. This dataset contains **1,460 residential home sale records** from Ames, Iowa, with **81 columns** (80 features + 1 target variable `SalePrice`). It includes a mix of numerical and categorical variables describing every aspect of the homes, and naturally contains missing values, outliers, and features suitable for engineering.

Required Libraries: `pandas`, `numpy`, `matplotlib`, `seaborn`, `scikit-learn`

Submission Guidelines

- Submit a single Jupyter Notebook (.ipynb) file.
- Each section must begin with a **markdown heading** matching the question number.
- All code cells must be **executed** with visible outputs (plots, tables, printed results).
- Include **written explanations** in markdown cells where indicated. Marks are awarded for both code and interpretation.
- Ensure your notebook runs top-to-bottom without errors.

Academic Integrity

This is an individual assignment. You may consult documentation, textbooks, and online tutorials, but all submitted code and writing must be your own work. Submissions will be checked for plagiarism. Clearly cite any external references used.

Deadline

Assignment needed to be submitted by 19th April 2026

Question 1 — Data Loading & Initial Exploration (10 marks)

1.1 Load and Inspect (4 marks)

1. Import all required libraries. **(1 mark)**
2. Load the `train.csv` dataset into a pandas DataFrame. **(1 mark)**
3. Display the first 5 rows and the shape of the dataset. **(1 mark)**
4. Use `.info()` to display the data types and non-null counts for all columns. **(1 mark)**

1.2 Summary Statistics (3 marks)

1. Generate descriptive statistics for all **numerical** columns using `.describe()`. **(1 mark)**
2. Generate value counts for any **three categorical** columns of your choice. **(1 mark)**
3. In a markdown cell, comment on any initial observations (e.g., large ranges, unexpected values, class imbalance in categories). **(1 mark)**

1.3 Data Type Assessment (3 marks)

1. Identify columns that have incorrect data types (e.g., `MSSubClass` is stored as an integer but represents a categorical dwelling type). List at least **two** such columns. **(1 mark)**
2. Convert these columns to the correct data types. **(1 mark)**
3. Explain in a markdown cell **why** correct data types matter for preprocessing and modelling. **(1 mark)**

Question 2 — Handling Missing Values (20 marks)

2.1 Missing Value Analysis (5 marks)

1. Calculate the number and percentage of missing values for **every column** that has at least one missing entry. Display the results as a sorted DataFrame. **(2 marks)**
2. Create a **heatmap** or **bar chart** that visually shows the missing data pattern across the dataset. **(2 marks)**
3. In a markdown cell, categorize the missing columns into groups: columns with a **high** (>40%), **moderate** (5–40%), and **low** (<5%) percentage of missing data. **(1 mark)**

2.2 Understanding the Missingness (5 marks)

1. For the columns `PoolQC`, `MiscFeature`, `Alley`, `Fence`, and `FireplaceQu`, explain in a markdown cell **why** these values are missing. Are they truly missing or do they represent the **absence** of a feature (e.g., no pool)? **(2 marks)**

2. For the column `LotFrontage`, investigate whether the missingness depends on the `Neighborhood` column by computing the median `LotFrontage` per neighbourhood. Display the result. **(2 marks)**
3. Classify the missingness mechanism for at least **three** columns as MCAR (Missing Completely at Random), MAR (Missing at Random), or MNAR (Missing Not at Random). Justify each classification in a markdown cell. **(1 mark)**

2.3 Imputation Strategies (10 marks)

Apply **different** imputation strategies based on the nature of the missingness:

1. For categorical columns where missing means "not applicable" (e.g., `PoolQC`, `Alley`, `Fence`), fill missing values with a label such as `"None"`. Show the code and verify with `.isnull().sum()`. **(2 marks)**
2. For `LotFrontage`, impute using the **median value grouped by `Neighborhood`**. Show the code and print the count of missing values before and after imputation. **(2 marks)**
3. For `GarageYrBlt`, impute using an appropriate strategy of your choice. Explain your reasoning in a markdown cell. **(2 marks)**
4. For any **one** numerical column with missing values, use `sklearn.impute.SimpleImputer` with the **mean** strategy. Show the code. **(2 marks)**
5. After all imputation is complete, verify that the dataset has **zero** remaining missing values. Display a summary to confirm. **(2 marks)**

Question 3 — Outlier Detection and Treatment (20 marks)

3.1 Visual Outlier Detection (6 marks)

1. Create **boxplots** for the following numerical columns: `SalePrice`, `LotArea`, `GrLivArea`, and `TotalBsmtSF`. Use a 2×2 subplot layout with clear titles and axis labels. **(3 marks)**
2. Create a **scatter plot** of `GrLivArea` (x-axis) vs. `SalePrice` (y-axis). Identify and annotate any visually obvious outliers on the plot. **(3 marks)**

3.2 Statistical Outlier Detection (8 marks)

1. **Z-Score Method:** Compute the Z-scores for `SalePrice` and `GrLivArea`. Identify all rows where the absolute Z-score exceeds 3 for either column. Display these rows and report how many outliers were found. **(3 marks)**
2. **IQR Method:** Compute the IQR for `LotArea`. Calculate the lower and upper bounds ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$). Identify all data points outside these bounds. Report the count. **(3 marks)**

3. In a markdown cell, **compare** the Z-score and IQR methods. Which one detected more outliers? Which method would you recommend for this dataset and why? **(2 marks)**

3.3 Outlier Treatment (6 marks)

1. Remove the two data points in `GrLivArea` that have values greater than 4000 sq ft but a `SalePrice` below \$300,000 (these are widely known outliers in this dataset). Show the scatter plot before and after removal. **(2 marks)**
2. For `LotArea`, apply **winsorization** (capping) at the 1st and 99th percentiles. Show the distribution before and after using histograms. **(2 marks)**
3. In a markdown cell, discuss when it is appropriate to **remove** outliers versus **cap/transform** them. Provide at least two considerations. **(2 marks)**

Question 4 — Feature Engineering (25 marks)

4.1 Creating New Features (10 marks)

Create the following new features and add them to the DataFrame:

1. `TotalSF` — the sum of `TotalBsmtSF`, `1stFlrSF`, and `2ndFlrSF` (total square footage). **(2 marks)**
2. `TotalBathrooms` — the sum of `FullBath`, `HalfBath`, `BsmtFullBath`, and `BsmtHalfBath` (count half-baths as 0.5). **(2 marks)**
3. `HouseAge` — calculated as `YrSold` minus `YearBuilt`. **(2 marks)**
4. `IsRemodeled` — a binary feature that is 1 if `YearRemodAdd` is different from `YearBuilt`, and 0 otherwise. **(2 marks)**
5. `TotalPorchSF` — the sum of `OpenPorchSF`, `EnclosedPorch`, `3SsnPorch`, and `ScreenPorch`. **(2 marks)**

4.2 Encoding Categorical Variables (8 marks)

1. Identify all remaining categorical columns in the dataset. Report how many there are. **(1 mark)**
2. For **ordinal** categorical columns (e.g., `ExterQual`, `BsmtQual`, `KitchenQual` which have values like Ex, Gd, TA, Fa, Po), apply **label encoding** using a custom mapping dictionary (e.g., `{ 'Po': 1, 'Fa': 2, 'TA': 3, 'Gd': 4, 'Ex': 5 }`). Apply this to at least **three** ordinal columns. **(3 marks)**
3. For **nominal** categorical columns (e.g., `Neighborhood`, `HouseStyle`), apply **one-hot encoding** using `pd.get_dummies()`. **(2 marks)**
4. In a markdown cell, explain the difference between **label encoding** and **one-hot encoding**, and describe when each is appropriate. **(2 marks)**

4.3 Feature Scaling (7 marks)

1. Select **five numerical features** of your choice and display their value ranges (min and max). **(1 mark)**
2. Apply **Min-Max Scaling** (using `sklearn.preprocessing.MinMaxScaler`) to these five features. Display the scaled ranges. **(2 marks)**
3. Apply **Standard Scaling** (using `sklearn.preprocessing.StandardScaler`) to the same five features. Display the mean and standard deviation after scaling. **(2 marks)**
4. In a markdown cell, explain when you would use Min-Max scaling versus Standard scaling. Give a practical example for each. **(2 marks)**

Question 5 — Data Visualization and Correlation Analysis (25 marks)

5.1 Distribution Analysis (7 marks)

1. Plot the **distribution** (histogram with KDE) of the target variable `SalePrice`. Comment on the skewness. **(2 marks)**
2. Apply a **log transformation** to `SalePrice` and plot the transformed distribution. Comment on how the shape changed. **(2 marks)**
3. Create a **grid of histograms** (at least 3×3) showing the distributions of nine numerical features of your choice. Use `plt.subplots()` with appropriate spacing. **(3 marks)**

5.2 Relationship Exploration (8 marks)

1. Create a **scatter plot matrix** (pairplot) for the following variables: `SalePrice`, `GrLivArea`, `TotalSF` (your engineered feature), `OverallQual`, and `HouseAge`. Use `seaborn.pairplot()`. **(3 marks)**
2. Create **boxplots** showing `SalePrice` grouped by `OverallQual` (quality rating 1–10). What trend do you observe? Write your answer in a markdown cell. **(3 marks)**
3. Create a **violin plot** or **swarm plot** showing `SalePrice` by `Neighborhood` for the **top 5** most common neighbourhoods. **(2 marks)**

5.3 Correlation Analysis (10 marks)

1. Compute the **Pearson correlation matrix** for all numerical features. **(1 mark)**
2. Create a **heatmap** of the correlation matrix. Use appropriate sizing, a diverging colour map, and display correlation values on the cells for the **top 15** features most correlated with `SalePrice`. **(3 marks)**
3. List the **top 10** features most positively correlated with `SalePrice` and the **top 5** most negatively correlated. Display as a sorted table. **(2 marks)**

4. Investigate **multicollinearity**: identify at least **two pairs** of independent features that are highly correlated with each other ($r > 0.8$). Explain in a markdown cell why multicollinearity can be a problem in regression models. **(2 marks)**
5. Compute the **Spearman rank correlation** between **OverallQual** and **SalePrice**. Compare it to the Pearson correlation and explain in a markdown cell why they might differ. **(2 marks)**